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Maryam Sana

Machine Learning Researcher

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A dedicated AI researcher with a strong background in deep learning, computer vision, and generative modeling, focused on solving complex real-world problems. Experienced in building robust and interpretable intelligent systems, and highly collaborative with the ability to communicate complex technical ideas clearly across interdisciplinary teams.

EDUCATION

Doctoral Candidate - Computational Neuroscience <i>Philipps-Universität Marburg</i>	2026 - 2029 Marburg , Germany
Masters of Computer Sciences and Engineering <i>CGPA: 4.0/4.3</i> <i>Kyung Hee University</i>	2023- 2026 Yongin-si , South Korea
Bachelor of Electrical Engineering <i>CGPA: 3.18/4.0</i> <i>National University of Sciences and Technology</i>	2017-2021 Islamabad, Pakistan

SKILLS

Programming	Python, C++, MATLAB, Bash, LaTeX
Machine Learning	PyTorch, TensorFlow, Hugging Face Transformers, Diffusers, NumPy, OpenCV
Neural Data & Signal Processing	MNE-Python, EEGLAB, EEG preprocessing and decoding, multi-subject neural datasets (THINGS-EEG2)
Real-Time & Deployment	TensorRT, INT8/FP16 quantization, NVIDIA Jetson, C++ inference optimization, low-latency model deployment
Infrastructure & MLOps	Git, GitHub Actions (CI/CD), Docker, DVC, Slurm GPU clusters, AWS EC2, GCP
Methods	Neural decoding, contrastive and representation learning, transformer encoders, probabilistic and generative modeling (VAEs, diffusion), information theory
Languages	English (C1), German (A1), Urdu (Native)

PUBLICATIONS

Perceptual Diffusion Video Compression with Motion-Aware Conditioning
Maryam sana, Hwi-Yong Kim, Jinwoo Choi
Keywords : [Video Compression, Diffusion Models, Human perceptualism, Generative AI]
Accepted : Summer Conference on The Korean Institute on Broadcasting and Media Engineers 2025

Generative Priors for EEG–Vision Representation Alignment
Maryam sana, Jose Garcia , Marco Rotherm , Hamidreza Jamalabadi
Keywords : [EEG Decoding Multimodal Representation Learning, Latent Space Alignment, Generative Models]
In Progress : Targetting ICLR 2027

PROFESSIONAL EXPERIENCE

Research Assistant / PhD Candidate <i>Psychcontrol Lab, University Hospital Giessen and Marburg (UKGM)</i>	Nov 2025 – Present Marburg, Germany
- Develop deep learning neural decoders for EEG-based visual decoding, aligning brain signals with CLIP visual–semantic embeddings for image retrieval and reconstruction. - Built an end-to-end EEG-to-image reconstruction pipeline combining a transformer-based EEG encoder, contrastive learning, VAE-based generation, and diffusion models. - Improved multi-subject decoding accuracy on THINGS-EEG2 using generative priors and hierarchical representation learning, raising retrieval from 22.6% to 25.2% top-1 and 50.8% to 56.7% top-5 over the ENIGMA baseline. - Processed and analyzed EEG recordings from 12 subjects (32 channels, 3000 trials) each, including preprocessing, artifact handling, and cross-subject evaluation. - Collaborate with neuroscientists and clinicians to design experiments and communicate results to both technical and non-technical audiences.	

Graduate Research Assistant

Kyung Hee University

Aug 2023 – Dec 2025

Yongin-si, South Korea

- Developed diffusion-based generative models for video frame prediction and compression, leading to **two first-author papers** (one accepted, one under review at IEEE Access).
- Integrated multimodal conditioning signals (optical flow and text) and quantified how auxiliary information reduces uncertainty in generative prediction.
- Benchmarked models on Vimeo, UVG, and HEVC datasets, reaching approximately **38 dB PSNR, 0.94 MS-SSIM, and 0.07 LPIPS** on UVG at 0.05 bpp while analyzing trade-offs between model complexity, bitrate, and perceptual quality.
- Analyzed how models allocate representational capacity across space and time, linking results to efficient coding and predictive processing principles from neuroscience.
- Trained large-scale models on **Slurm-managed GPU clusters** and built reproducible experiment pipelines.

Machine Learning Engineer

Lastmile AI

Aug 2022 – Jun 2023

Remote

- Built a named entity recognition pipeline for payslips, bank statements, and passports by fine-tuning [LayoutLMv2](#), a transformer-based document model, reaching **95% precision**.
- Reduced inference latency to **under one second** using C++ inference APIs and model quantization.
- Set up CI/CD with **GitHub Actions** for automated testing and deployment on AWS EC2.

Software Engineer

Confiz LLC

Jul 2021 – Aug 2022

Lahore, Pakistan

- Built a React/Redux web application and RESTful services backed by Azure databases to support supply chain planning for a large grocery retail chain.

Undergraduate Research Assistant

TU Kaiserslautern Lab, NUST

Aug 2019 – Jun 2021

Islamabad, Pakistan

- Built a **real-time** disaster response system combining image classification and object detection to localize victims and vehicles in aerial imagery.
- Deployed the full pipeline on **NVIDIA Jetson** with **TensorRT** and **INT8/FP16 quantization**, raising throughput from **9 FPS to over 50 FPS** with minimal accuracy loss under strict latency and power constraints.
- Evaluated trade-offs between inference latency, memory footprint, power consumption, and accuracy to validate deployment on edge devices.
- Studied binarization and quantization of CNNs (ResNet, LeNet, Inception, YOLO) following [QNNs \(Hubara et al.\)](#) for efficient inference on constrained hardware.

AWARDS AND FUNDING

- 2025 - Present Grant by National Research Foundation of Korea in "**Towards Generalizable Video Understanding: Learning Debaised Video Representations**"
- 2024 - Grant by Institute of Information & Communications Technology Planning and Evaluation in "**AI-based OTT user and Content data analysis and video recommendation system development**"
- 2019 - DAAD grant for "**Early Detection of Forest Health Decline using Remote Sensing**", German Academic Exchange Service (DAAD)
- 2025 - Summer Conference Outstanding Papers for "**Improving Video Recognition Quality through a deep learning-based preprocessing framework.**"